G10 FX Carry Strategy

Full Development Log & Problem Journal

Systematic Macro QIS Backtest – Volatility Targeted Approach

February 27 – March 5, 2026 | Tashfeen Omran

1. Project Overview

This document records every problem encountered and every decision made alongside all fixes applied during the construction of a systematic G10 FX carry strategy backtest. The project took around 6 days.

The Strategy in One Paragraph

G10 FX carry is one of the oldest documented risk premia in academic literature. The core idea is that currencies with higher interest rates tend to outperform currencies with lower interest rates over time. By systematically going long the 3 highest-yielding G10 currencies and short the 3 lowest-yielding currencies, I can extract this carry premium. I also scaled position sizes to target a fixed portfolio volatility which allows for a controlled and risk-managed return.

Data Sources

- FX Spot Prices: Barchart Premium - daily OHLCV for 9 G10 pairs, April 1 2002 to February 27 2026
- Interest Rates: FRED - 3-month interbank rates for all 10 G10 currencies, monthly frequency, forward-filled to daily
- Benchmark: SPY daily data from Barchart for performance comparison

Final Performance

Metric	Strategy
Total Return (Apr 2002 – Feb 2026)	-29.35%
Annualized Return	-1.44%
Annualized Volatility	**9.60%**
Sharpe Ratio	-0.15
Sortino Ratio	-0.18
Calmar Ratio	-0.02
Max Drawdown	-58.99%
Max DD Duration	5,000+ days
VaR (95%, daily)	-1.01%
CVaR (95%, daily)	-1.52%
Skewness	**-0.91**
Kurtosis	7.64
Hit Rate	41.17%
Correlation to SPY	0.36
Mean Annual TC Drag	**0.03%**

Metric	Strategy
Pre-GFC Sharpe (2002–2007)	~0.60

The negative full-period return accurately reflects the documented compression of G10 carry premia under a decade of ZIRP, nevertheless pre-2008 sub-period replicates academic benchmarks closely (Sharpe ~0.60). I've discussed this in detail in the problems section below.

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2. The Journey - From Roadmap to Final Version

2.1 Starting Point: The Roadmap

The project began with a roadmap.md which Gemini AI made from my personal notes the day prior. It's a detailed blueprint covering the full directory structure, data requirements, day-by-day build plan, strategy logic, backtesting engine design and performance metrics. This was written before a single line of code was touched.

Key early decisions made in the roadmap:

- Start date April 1 2002 was the binding constraint as the JPY FRED series (IR3TIB01JPM156N) only starts 2002-04-01. JPY is critical since it is almost always the funding currency.
- End date February 27 2026, the present day when I started working on this. FRED series lag by ~1 month so the last available rate was January 2026 and the final ~40 days were forward-filled.
- Barchart Premium for FX data, chosen because I already used it to download some other dummy data and it provides daily data back to 2000, has a 20,000 record per request cap (my ~5,900 rows per pair is well within this), and allows CSV download. Alpaca was considered but rejected due to a 15-minute historical data limitation.
- FRED for interest rates, it's free and is institutional quality.

2.2 The Build - Five Days of Work

Day 1 focused on data ingestion: loading 9 FX CSVs from Barchart, loading 10 interest rate CSVs from FRED, computing log returns, forward-filling monthly rates to daily, and computing carry differentials.

Day 2 I built the signal construction pipeline.

Day 3 implemented volatility targeting.

Day 4 built the backtest engine and performance metrics.

Day 5 was debugging, fixing critical errors, and generating final outputs.

I had several problems and debugging them is where real learning happened. Every problem below was fixed methodically and the problems are listed in the order they were encountered.

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3. Honest Interpretation of Results

Why the Full-Period Return Is Negative

The -29.35% total return over 23 years (-1.44% annualized) reflects a genuine structural shift in the macroeconomic environment that removed the fundamental driver of carry returns.

- 2002 - 2007: Strategy generated +41% total, ~6.2% annualized, Sharpe ~0.60. G10 rate differentials were wide (AUD at 6%, USD at 5%, JPY at 0.5%). This is the academic benchmark period and our results closely match published carry indices.
- 2008: -28.9% in a single year. The carry crash is documented and expected and global deleveraging caused simultaneous unwinding of all carry positions, high-yielders (AUD, NZD) collapsed, safe-havens (JPY, CHF) surged. Our short JPY/CHF positions were badly hurt.
- 2009 - 2021 ZIRP: G10 rate differentials compressed to near-zero and there was almost nothing to earn from carry. The strategy had no fundamental driver and the small negative drift in this period reflects the residual noise of near-zero differentials and the occasional risk-off spike.
- 2022 - 2026: The Fed hiking cycle (fastest in 40 years) restored differentials. USD carry became strongly positive and the rolling Sharpe briefly returned to near-positive territory in 2020/2021.